

Final Report & Literature Review

Group 17:

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I. INTRODUCTION

Semantic segmentation is a computer vision tool with the task of assigning a semantic label to each pixel of an image, or in other terms, defining boundaries between different semantic classes. [?] Semantic segmentation has been successfully applied in numerous different contexts, from medical imaging to self-driving cars [?], [?].

In 2020 NASA has published an open dataset of images with segmentation masks from multiple Mars missions, taken by rovers currently operating on Mars[?]. Numerous groups have evaluated the capacity of different neural networks in executing semantic segmentation tasks on the AI4Mars and similar dataset, including access time to prediction in the limited hardware of the rovers [?], [?], [?].

In this work we focus our effort into evaluating the choice of loss functions in image segmentation tasks in the m2020 navigational dataset from the AI4Mars dataset, containing 2355 images and masks. We also explore the effects of pre-training the model of choice with the MSL Navigational dataset, containing 16396 labeled images, a retraining strategy that provided mixed results in previous works [?, swan2021]

The model chosen was the DeepLabV3+ with a ResNet-50 backbone, a model that provides time to prediction of 25ms and mean accuracy of 87% when trained and tested in a similar dataset of martian terrain images by Luna et al [?, Luna2025]n an independent dataset of Mars terrain images. We apply this model to the open AI4Mars dataset to evaluate if the pre training with the MSL and the choice of loss functions can improve the recall and mean Intersection over Union (mIoU) results in comparison to each other, and related work.

In section 2 we review relevant previous work, both on categorizing different models for semantic segmentation, as well as previous results of other attempts to do image segmentation on the AI4Mars or similar independent datasets. In section 3 we present our strategy to process the data and train the model with two different loss functions - Cross entropy and Log Cosh Dice Loss - with and without pre training on the MSL dataset. In section 4, we present the results, and discuss them in relation to other relevant works.

II. LITERATURE REVIEW

A. Semantic Segmentation

Models:

Loss Functions:

B. AI4Mars

III. TRAINING METHOD

A. Dataset and Preprocessing

The m2020 navigational dataset, consisting of 2355 images with class label masks. The masks were produced by a crowd science initiative that trained labelers to identify the four classes: soil, bedrock, sand and big rocks, represented by pixel color values of 0,1,2,3 respectively. The masks produced by the trained labelers was reviewed and finalized by a group of expert labelers to produce the final dataset. The images and masks aren't always the same size, some images were centercropped before the labels were constructed, for this images it's necessary to center crop the images accordingly when retrieving them from the dataset [?].

For training, The dataset is split randomly into 90% for the training set and 10% for the validation set. Both datasets are resized, grayscaled, and converted into normalized tensors using the mean and variation suggested by the PyTorch implementation of the DeepLabV3+ model [?], with the mean by RGB channel as $\text{mean}=[0.485, 0.456, 0.406]$, with standard deviation of $[0.229, 0.224, 0.225]$.

The training dataset additionally passes through an augmentation loop which includes:

- Horizontal flip (50% probability)
- Rotation up to 20° (50% probability)
- Random cutouts (up to 8, 50% probability)
- Random brightness adjustment (20% probability)

One recurring problem in related works [?], [?] was that big rocks are underrepresented, consisting of only 3.2% of non-background pixels as shown in figure 1, to compensate a little bit for this during training, the data-loader was set with a weighted sampler that favored loading images with big rock pixels in their mask by a factor of 15.

B. Model Training: DeepLabV3+ with ResNet-50

DeepLabV3+ belongs to a family of neural networks that exploits dilated (atrous) convolutions to increase the field of view without increasing the number of parameters. While Luna et al [?] was successful in implementing DeepLabV3+ on an independent dataset of Martian images, this work implements DeepLabV3+ with a ResNet-50 backbone on the open AI4Mars dataset.

We utilized the PyTorch implementation of DeepLabV3+, which employs Atrous Spatial Pyramid Pooling (ASPP) with an encoder-decoder topology [?]. The training process

uses different initial learning rates: 0.0005 for the classifier (ASPP onwards) and 0.00001 for the ResNet-50 backbone.

The optimizer is Stochastic Gradient Descent (SGD) with a decay rate of 0.0001 and momentum of 0.9. A polynomial scheduler is used to dynamically decrease the learning rate with a power of 0.9.

C. Training Procedure

The model is trained for a total of 150 epochs using a progressive resizing strategy:

- **Epochs 1–50:** 128×128 pixels resolution
- **Epochs 51–100:** 256×256 pixels resolution
- **Epochs 101–150:** 512×512 pixels resolution

The model was trained with two possible choices of loss function: weighted Cross Entropy Loss, and weighted Log-Cosh Dice Loss. The cross entropy loss scores the model in its ability to correctly label each pixel, while the Log-Cosh Dice Loss scores based on the ability to adequately define the correct boundaries of a region. The choice of the log cosh version of dice loss is due to the big rock class often being present in just a small section of the image, resulting in fluctuating values for the pure dice loss. The choice of log-cosh dice loss makes the dice loss a bit more resistant to these variations [?].

The loss functions was also weighted inversely with the abundance of each class, according to figure 1, so the soil class was weighted as for 4% of the final loss value, 8% for bedrock, 15% for sand and 73% for big rock. The loss functions also ignored the background, which in retrospect was a mistake, the final models learned to always guess a class (usually soil) to minimize their loss, since “missing” the background isn’t penalized, as shown in figures 4 and 5.

For each choice of loss function, the model either trained directly from the default weights of the torch implementation of DeepLabV3+, or it was initialized from the default weights, then pre trained on the 16396 images of the MSL dataset and then trained and access on the m2020 datasets. This training process resulted in four different final models with different scores

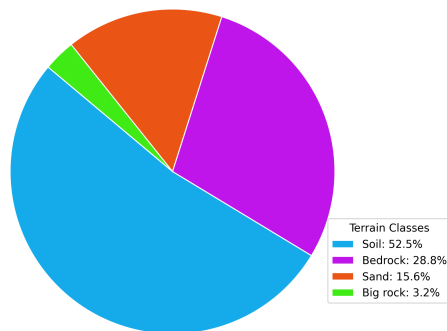


Fig. 1: Relative abundance of each non-background class in the AI4Mars m2020 navigational dataset.

IV. RESULTS

The evolution of the assessment on accuracy and mean intersection over union (mIoU) of the four versions of the models with different trained set up are presented in figures 2 and 3. The cross entropy loss trained models had their training accelerated by the resizing strategy, with the accuracy quickly increasing after the image size. The pretraining strategy didn’t significantly improve the accuracy and mIoU of the cross entropy based model, with the model achieving an accuracy of around 80.4% and mIoU of 64% regardless of pretraining, this is consistent with the finding of previous works on the AI4Mars dataset [?]. The findings for the log cosh Dice Loss trained model is similar in the overall assessment, with maximum accuracy of around 70.7% and mIoU of 60% regardless of pretraining.

On the ability to classify specific classes, the cross entropy loss trained models seem to outperform the log cosh Dice Loss model significantly in the underrepresented big rock class, achieving around 74% accuracy and recall on that class, against 57% on these metrics for the best performing dice loss based models. In the dice loss based models, pretraining HURT the ability of the model in identifying big rocks, with the accuracy reducing from 57% to 49% with pretraining.

It’s important to also note a problem in the choice of ignoring the background when computing this losses, due to this choice, the model learning to ALWAYS guess a class for each pixel, as shown in figures 4 and 5. The prediction don’t contain any background prediction (colored white on the ground truth)

The work by Luna et al [?] also trained the DeepLabV3+ with ResNet-50 as a backbone on an independent but similar dataset to the AI4Mars dataset. Their model achieved 87% accuracy and 72% mIoU. It’s not likely we can know where their improvements can from, if it was due to better choice of loss function and training hyperparameters, or if it was due to the different dataset.

Example of predictions for the best performing Log Cosh Dice Loss based model is shown in figure 4, and for the best performing cross entropy model...

The results per class assessed on Recall and IoU are presented in table [?]. The work by Atha et al [?] - presented on table [?] trained models with ResNet-101 backbone, serving as a point of comparison for our results, specially in investigating the effects of the additional DeepLabV3+ layers. Our training methods seemed to have universally improved the prediction of soil, but that is most likely due to our model ignoring the loss on predicting the background, this resulted in the model always guessing something for every pixel, and soil was the most common class. That said, the Recall results for the other classes also shown an improvement over the pure ResNet-101, and for some classes, and improvement over the ResNet-101 with pretraining. The ResNet-101 with ImageNet still outperforms DeepLabV3+ with ResNet-50 back bone.

Metric	Class	ResNet-101	+ImageNet	+MSL
Precision	Soil	0.933	0.999	0.916
	Bedrock	0.857	0.467	0.917
	Sand	0.664	0.667	0.754
	Bigrock	0.558	0.618	0.174
Recall	Soil	0.746	0.596	0.838
	Bedrock	0.570	0.958	0.536
	Sand	0.975	0.952	0.957
	Bigrock	0.283	0.355	0.640

TABLE I: Performance metrics for models trained on the M2020 dataset by Atha et al. [?], reorganized for single-column fit.

Metric	Class	CE	CE+PT	LCDL	LCDL+PT
Recall	Soil	0.871	0.864	0.899	0.923
	Bedrock	0.825	0.853	0.773	0.725
	Sand	0.811	0.762	0.587	0.695
	Bigrock	0.710	0.743	0.572	0.496
IoU	Soil	0.789	0.790	0.712	0.728
	Bedrock	0.725	0.753	0.683	0.651
	Sand	0.668	0.586	0.498	0.605
	Bigrock	0.412	0.443	0.490	0.434

TABLE II: Recall and IoU metrics for the m2020 validation dataset using DeepLabV3+ with ResNet-50 across various configurations. CE: Cross Entropy Loss, LCDL: Log Cosh Dice Loss, PT: Pretrained.

V. CONCLUSION

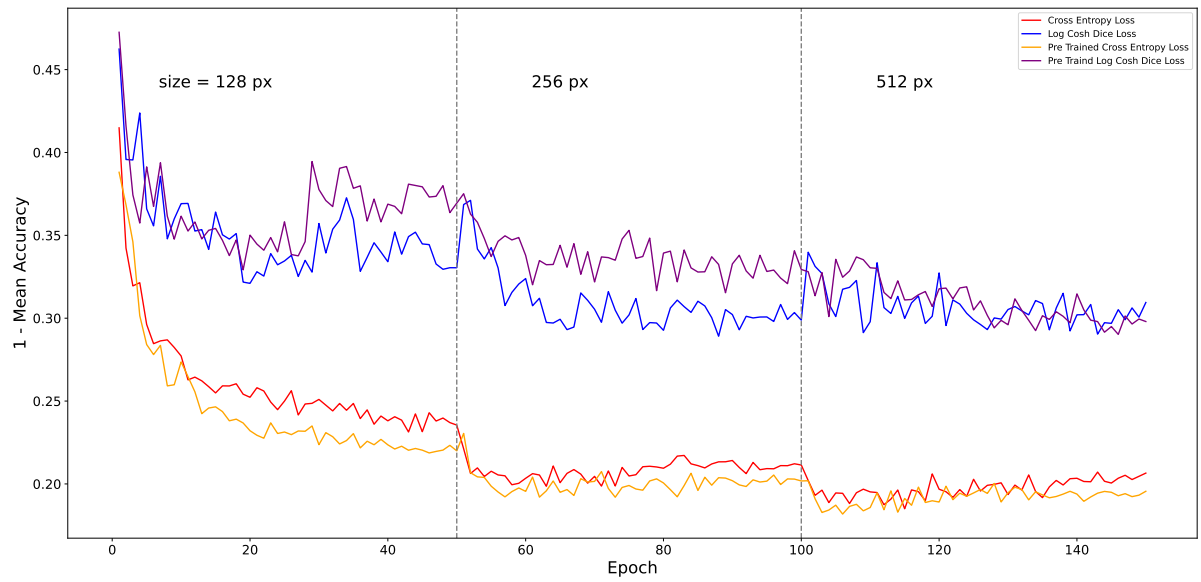


Fig. 2: Evolution of 1 - accuracy during training. At each 50 epochs, the model was trained with larger images

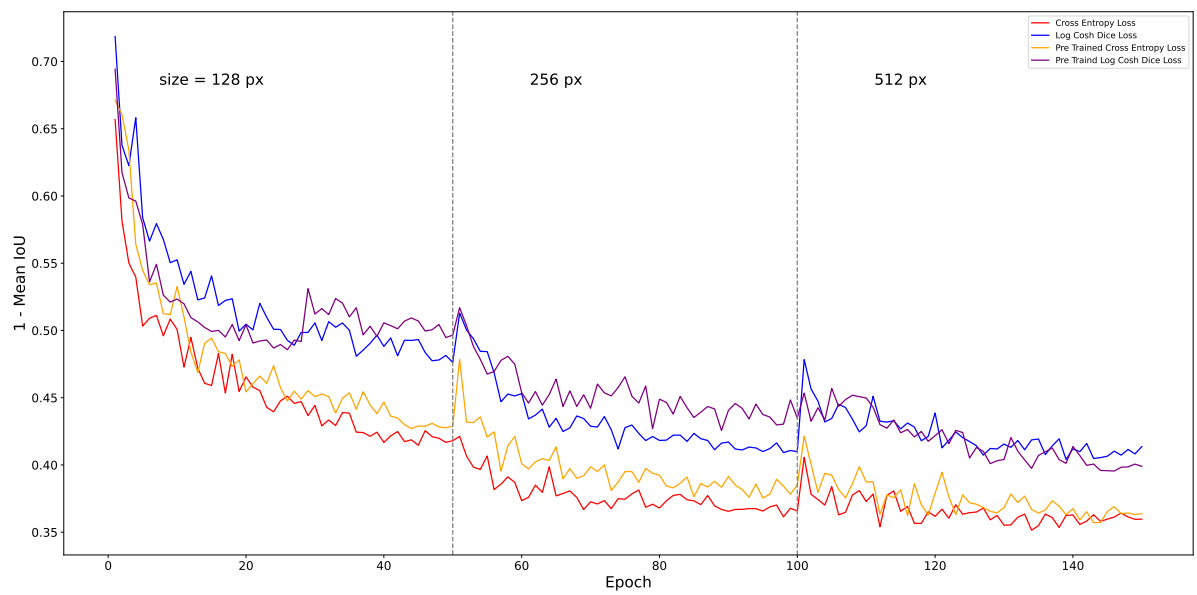


Fig. 3: Evolution of 1 - Mean Intersection over union during training. At each 50 epochs, the model was trained with larger images.

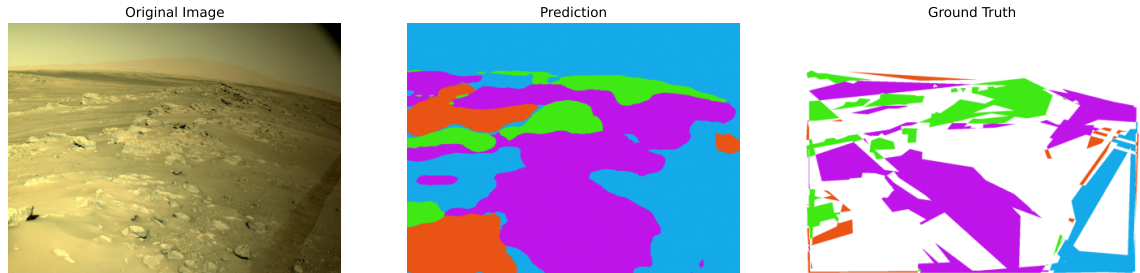


Fig. 4: Example of original image, model prediction and ground truth mask for the best performing model trained with the Log Cosh Dice Loss function. This model was pretrained on the MSL dataset. The label colors are: Blue for soil, Purple for Bedrock, Orange for Sand and Green for Big Rock

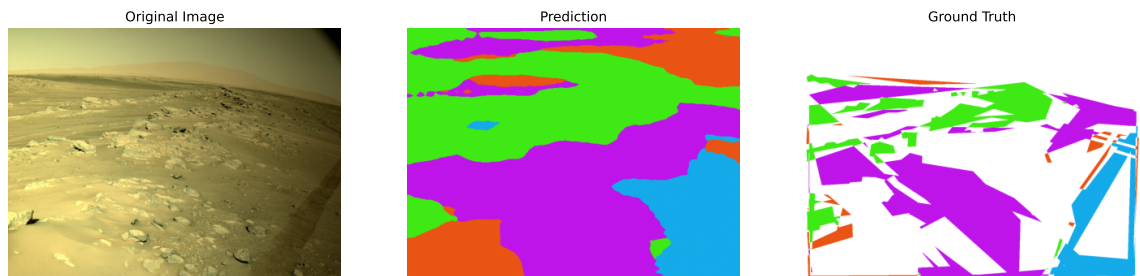


Fig. 5: Example of original image, model prediction and ground truth mask for the best performing model trained with the Cross Entropy Loss function. This model was pretrained on the MSL dataset. The label colors are: Blue for soil, Purple for Bedrock, Orange for Sand and Green for Big Rock